

Association between antiviral treatments and fracture in elderly patients with HBV needs further evaluation

To the Editor:

We read with great interest the article titled 'Long-term use of tenofovir disoproxil fumarate increases fracture risk in elderly patients with chronic hepatitis B' published in a recent issue of the *Journal of Hepatology*.¹ The topic is particularly intriguing as it explores the impact of entecavir (ETV) and tenofovir disoproxil (TDF) on the risk of bone fracture in patients with chronic hepatitis B based on a real-world territory-wide cohort. Their findings revealed that long-term TDF treatment for 2 years or more is associated with a higher incidence of bone fracture in older patients. Although the study was elegantly designed and inspiring, we believe that certain aspects warrant further discussion.

Firstly, the inclusion criteria of this study did not clearly specify whether the enrolled patients exhibited traits related to osteoporosis or sarcopenia. It is crucial to note that osteosarcopenia is a common complication among patients with chronic liver disease, exhibiting a significantly higher prevalence among those with cirrhosis compared to those without.^{2,3} Additionally, individuals with osteosarcopenia are significantly more likely to experience falls, fractures, and mortality than in non-osteosarcopenic groups. Therefore, this information is imperative for the readers' knowledge, and baseline variables which should include bone mineral density and sarcopenia index prior to the antiviral treatments are needed to draw a robust conclusion.

Secondly, we noted that TDF-treated patients with a match were older and more likely to be female, as per the propensity score-matching. The authors discovered that patients receiving TDF had a higher risk of developing incident fractures compared to those treated with ETV after 24 months (weighted subdistribution hazard ratio 2.33, 95% CI 1.20–4.55, $p = 0.014$ for age >65 years; weighted subdistribution hazard ratio 2.12, 95% CI 1.01–4.46, $p = 0.047$ for age >70 years).¹ Similarly, age and sex could both be risk factors for osteoporosis and fracture, which occur as part of the natural aging process, particularly in relation to menopause and the alterations in sex hormones that accompany advancing age.^{4,5} Additionally, the intersection of the cumulative risk curves between the two groups in Fig. 3B should not be ignored. Importantly, the conventional method of using the competing risk model and hazard ratio for assessing the long-term treatment outcome can be suboptimal in terms of interpretation and power in this study.⁶ Restricted mean survival time is suggested to quantify the treatment effect and has been discussed extensively in

survival analysis.⁷ Moreover, the process of calculating the difference in two restricted mean survival times is always valid without any model assumptions.⁸ We suggest that if the authors added more comprehensive statistical methods into their subgroup analysis, it may solve this problem.

Thirdly, the study did not investigate the impact of switching therapies. It would be helpful to gain insight into the management of patients during the transition of antiviral therapy. For instance, evaluating whether there is a decline in the frequency of fractures in patients who initially were prescribed TDF but later switched to ETV or tenofovir alafenamide.

Fourthly, previous studies have demonstrated that calcium intake and selenium levels can impact bone mineral density at specific skeletal sites and are associated with bone metabolism and remodeling.⁹ Moreover, physical activity is an important modifiable risk factor for osteoporosis and has the potential to mitigate the risk of hip fractures in older women, although the specific type and duration of activity necessary have yet to be established.¹⁰ In order to further validate whether long-term use of TDF increases the risk of fractures in elderly patients with chronic hepatitis B, it is advisable to ensure consistency in terms of physical activity intensity and diet composition between the two groups.

In conclusion, while acknowledging the aforementioned points that merit careful consideration, we are genuinely grateful for the contribution of Yip *et al.* in their work, and their findings support the current treatment guidelines, advocating for the customization of antiviral therapy based on age and comorbidities.

Jia-Lin Wu
Jun-Yang Luo
Zai-Bo Jiang*

Department of Interventional Radiology, The Third Affiliated Hospital of Sun Yat-sen University, Guangzhou 510630, China

*Corresponding author. Address: Department of Interventional Radiology, The Third Affiliated Hospital, Sun Yat-sen University, 600 Tianhe Road, Guangzhou 510630, China.

E-mail address: jiangzbo@mail.sysu.edu.cn (Z.-B. Jiang)

Received 9 February 2024; Received in revised form 8 March 2024;

Accepted 11 March 2024; Available online 26 March 2024

<https://doi.org/10.1016/j.jhep.2024.03.028>

© 2024 European Association for the Study of the Liver. Published by Elsevier B.V. All rights reserved.



Financial support

The authors received no financial support to produce this manuscript.

Conflict of interest

The authors declare that they have no competing interests.

Please refer to the accompanying ICMJE disclosure forms for further details.



ELSEVIER

Authors' contributions

All authors made significant contributions to the intellectual content of the manuscript and approved its final version.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.03.028>.

References

- Author names in bold designate shared co-first authorship*
- [1] **Yip TC-F, Lai JC-T**, Yam T-F, et al. Long-term use of tenofovir disoproxil fumarate increases fracture risk in elderly patients with chronic hepatitis B. *J Hepatol* 2024;80:553–563.
 - [2] Ebadi M, Bhanji RA, Mazurak VC, et al. Sarcopenia in cirrhosis: from pathogenesis to interventions. *J Gastroenterol* 2019;54:845–859.
 - [3] Guañabens N, Parés A. Osteoporosis in chronic liver disease. *Liver Int* 2018;38:776–785.
 - [4] Kanis JA, Cooper C, Rizzoli R, et al. European guidance for the diagnosis and management of osteoporosis in postmenopausal women. *Osteoporos Int* 2019;30.
 - [5] Osteoporosis Prevention, Screening, and diagnosis: ACOG clinical practice guideline No. 1. *Obstet Gynecol* 2021;138:494–506.
 - [6] Trinquart L, Jacot J, Conner SC, et al. Comparison of treatment effects measured by the hazard ratio and by the ratio of restricted mean survival times in oncology randomized controlled trials. *J Clin Oncol* 2016;34:1813–1819.
 - [7] **Horiguchi M, Tian L, Uno H**, et al. Quantification of long-term survival benefit in a comparative oncology clinical study. *JAMA Oncol* 2018;4:881–882.
 - [8] Pak K, Uno H, Kim DH, et al. Interpretability of cancer clinical trial results using restricted mean survival time as an alternative to the hazard ratio. *JAMA Oncol* 2017;3:1692–1696.
 - [9] **Qu Z, Yang F**, Yan Y, et al. Relationship between serum nutritional factors and bone mineral density: a mendelian randomization study. *J Clin Endocrinol Metab* 2021;106:e2434–e2443.
 - [10] Gregg EW, Cauley JA, Seeley DG, et al. Physical activity and osteoporotic fracture risk in older women. Study of Osteoporotic Fractures Research Group. *Ann Intern Med* 1998;129:81–88.